

COMP167 Lab 8 - Interfaces

You will be implementing the Comparable and the Comparator interfaces in this lab.

Card class (5pts):

First, create a class named Card with the following properties: color:String, value:String. The card would be used in the game of Uno. Add two constructors (no-arg constructor and a constructor that includes both properties), toString(), mutator and accessor methods. The toString method should use the “,” to separate properties.

Your Card class should implement the Comparable interface as discussed in class. Implement the compareTo method that sorts by color then by value. So, if the cards have the same color, you need to look at the value to determine whether one card is less than the other. Use the String *compareTo* method to compare colors and values.

Deck Class:

Create a Deck class consisting of 76 Cards represented in an ArrayList<Card> named cards. There are 0 x 1, 1 to 9 x 2 for each color (Red, Green, Blue, Yellow) = 76 cards. The game of Uno has other cards; however, we are ignoring them for this exercise. Here is an example of creating just the Green cards with the number values set that would be a part of the Deck constructor:

```
cards.add( new Card( "green", "0" );
for (int i=1; i<=9; i++) {
    cards.add( new Card( "green", i+"") );
    cards.add( new Card( "green", i+"") );
}
```

void shuffle() - Add a method to your Deck class that shuffles (randomizes) the cards. This method should loop through the entire deck of 76 cards and assign that card to another random location within the deck. Repeat this loop 7 times (so you will have nested loops where the outer loop is executed 7 times and the inner loop is executed 76 times). Use the java.util.Random class for your random number generator.

void sort() - This method should sort your deck and should simply consist of a call to the static method sort() of the Collections class. This Collections.sort() method will sort your ArrayList since your Card class implements the Comparable interface.

String toString() – return a string with all the cards separated by a “|”. Insert a newline after every 15 cards.

Main Method:

- Your main method should :
- Instantiate a new deck of cards
- Shuffle the cards
- Output the shuffled (randomized) cards to a JOptionPane.
- Sort the cards.
- Output the sorted cards to a JOptionPane.

Comparator:

Create a class named ValueComparator that implements the Comparator interface in a separate file. This comparator should compare Card objects by their value only. Add a new valueSort() method to your deck class. This method should create an instance of the ValueComparator class and then pass that instance as the second parameter to the overloaded Collections.sort method. Test your new method by sorting your deck of cards by value only in your main() method.

Grading:

- Implement the Card class (5pts).
- Implement the Deck constructor and toString() method. The constructor populates the ArrayList with 76 cards (6pts).
- Implement the Deck shuffle() method (7pts).
- Implement the Deck sort method that uses the Comparable interface (7pts).